



ONAMA

A Quantum Leap in HPC

This article introduces readers to Onama, a CDAC-created bundle of well-selected parallel and serial applications for various engineering disciplines. An amalgamation of HPC and open source technologies, it is based on Linux, and aims to empower engineers with powerful and effective tools, encouraging them to invent and innovate at a rapid rate.

“Scientists investigate that which already is; engineers create that which has never been.” - Albert Einstein

As Albert Einstein rightly said, engineering is the practical approach of humans to develop new machines, mechanisms and processes. Throughout human history, engineers have been responsible for rapid technological developments, be it fundamental inventions such as the wheel, or present-day high-tech lasers, spaceships, and supercomputers. Engineering has penetrated into every possible sector, and has helped make our lives easier, safer, and more comfortable than those of our forefathers. And through these several achievements, engineers seek only one objective: technological utopia. This dream can be achieved only by empowering our engineers with powerful and effective tools, and encouraging them to come up with inventions and innovations at a rapid rate. Thus, much effort has gone into the research and production of cutting-edge tools and equipment to help engineers reach this goal. One significant development

of our times is the application of high-performance computing.

High-performance computing, also known as HPC, is the use of supercomputers and computer clusters in existing computing systems. It is expected that, in the coming years, HPC will help us catch up with the growing need of faster and better computers in all sectors. To meet this anticipated demand, C-DAC has created an effective and versatile solution for today's engineers and engineering students, called Onama. This is an integrated package comprising a well-selected set of parallel as well as serial applications, to be used across various engineering disciplines. It is an amalgamation of HPC and open-source tools and technologies, and operates through the popular Linux operating system. It contains numerous nVIDIA CUDA-enabled applications that are widely used in domains such as molecular dynamics and bio-informatics. Figure 1 shows the hierarchical diagram of Onama's